

ME 612: Introduction to Bio-materials

Course Code	ME 612
Course Name	Introduction to Bio-materials
L-T-P-C	3-0-0-3
Intended for	B.Tech., M. Tech., M.S., and Ph. D.
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Objectives

Developing materials for use in medicine is a challenging interdisciplinary process and requires an understanding of material bulk and surface properties, the various biological responses to the materials, the clinical context of their use, manufacturing processes, cost, sterilization, packaging and regulatory issues. This course is designed to introduce students to the various classes of biomaterials in use and their application in selected subspecialties of medicine.

Upon successful completion of this course, the student will be able to:

1. Differentiate the various classes of biomaterials on the basis of structure and function;
2. Differentiate various analytical methods based on their use to characterize bulk and surface properties of biomaterials
3. Differentiate the molecular and cellular events that follow exposure of materials to bodily fluids and to contact with various tissues of the human body;
4. Differentiate various biomedical devices based upon function, biomaterial composition, patient risk and clinical application; and to
5. Describe various practical aspects of biomedical device design, fabrication and testing.

Course Contents

Introduction to basic concepts of Materials Science; Salient properties of important material classes; Property requirement of biomaterials; Concept of biocompatibility; Structure and properties of biological cells & tissues; cell-material interactions and foreign body response; Assessment of biocompatibility of biomaterials, in vitro biochemical assays (cellular adhesion, cellular viability using MTT, osteogenic differentiation using ALP assay; Biomaterialisation using Osteocalcin assay);

In vivo testing and histocompatibility assessment; genotoxicity assessment (Physical damage to DNA by biomaterial eluates); important biometallic alloys: Ti-based, stainless steels, Co-Cr—Mo alloys; Bioinert, Bioactive and bioresorbable ceramics; Processing and properties of different bioceramic materials with emphasis on hydroxyapatite; synthesis of biocompatible coatings on structural implant materials; plasma spraying of carbon nanotube reinforced hydroxyapatite on Ti-6Al-4V substrate; Microstructure and properties of glass-ceramics; biodegradable polymers; Design concept of developing new materials for bio-implant applications.

References and text books

1. Ratner, Hoffman, Schoet and Lemons, (Editors), **Biomaterials Science: An introduction to Materials in Medicine**, 2nd Edition, Elsevier Academic Press, 2004.
2. Ritchie and Karihalo, **Comprehensive structural integrity**, Vol.:9 Bioengineering, Elsevier, Academic Press, 2003.
3. Fredrick H. Silver and David L. Christiansen, **Biomaterials Science and Biocompatibility**, Springer, New Jersey.
4. Janathan Black, **Biological Performance of Materials: Fundamentals of Biocompatibility**, Marcel Dekker, 1981.
5. J.M. Davis (Editor), **Basic Cell Culture: A Practical Approach**, IRL Press, Oxford University Press, 1994.